

*Part A***Question 1:**

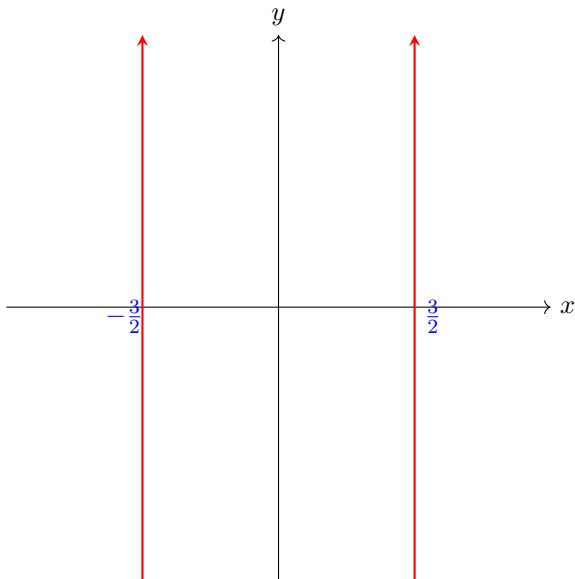
$$A = \{(x, y) \in \mathbb{R}^2 : 2|x| \leq 3\}$$

$$B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$$

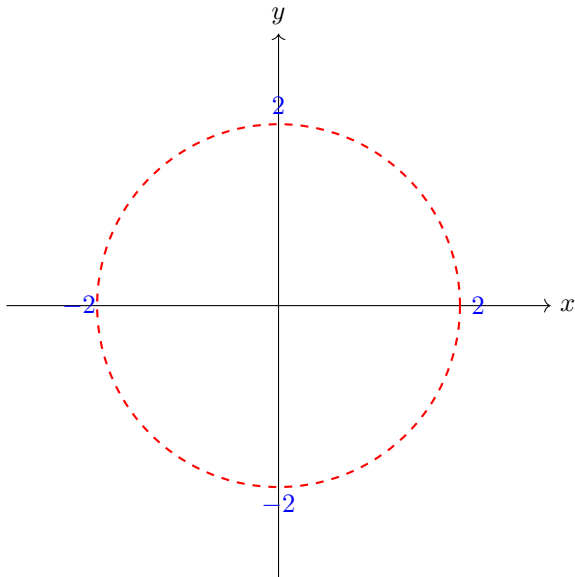
$$C = A \cap B$$

**a)**

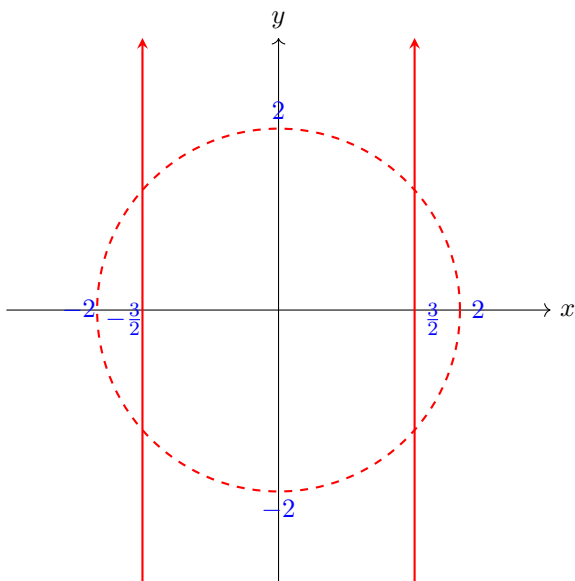
Sketch of set (A):



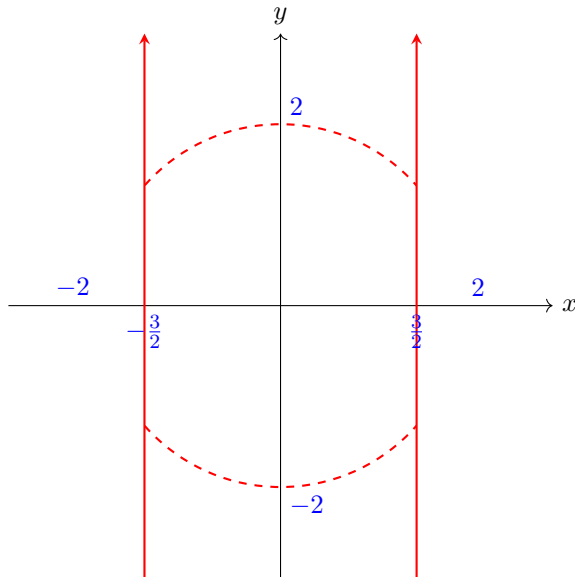
Sketch of set (B):



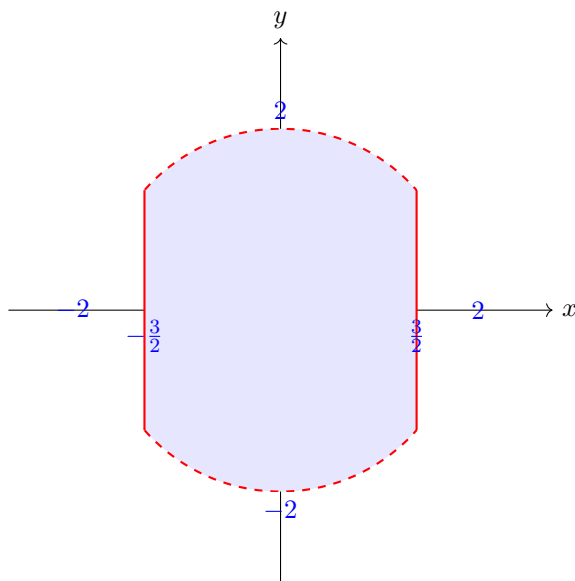
Sketch of set (C):



Maybe



Maybe 2



**b)**

**i.**

By Definition 3.6 HB Chapter 16 p84 ,

let  $x = \frac{3}{2}$  and solve for  $y$  in set B's inequality:

$$x^2 + y^2 < 4$$

;

then  $y = \sqrt{2^2 - (\frac{3}{2})^2} = \sqrt{\frac{7}{4}} = \frac{\sqrt{7}}{2}$ . Thus, one closure point is  $(\frac{3}{2}, \frac{\sqrt{7}}{2})$ .

ii.

Closure of set C is:

$$C1 = \{(x, y) \in \mathbb{R}^2 : 2|x| \leq 3 \text{ and } x^2 + y^2 \leq 4\}$$

Interior of set C is:

$$Int = \{(x, y) \in \mathbb{R}^2 : 2|x| < 3 \text{ and } x^2 + y^2 < 4\}$$

Boundary of set C is:

$$Bd = \{(x, y) \in \mathbb{R}^2 : (2|x| = 3 \text{ and } x^2 + y^2 \leq 4) \text{ or } (2|x| \leq 3 \text{ and } x^2 + y^2 = 4)\}$$

c)

Given the 'mixed' metric  $d$ ;

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + d_0(y_1, y_2)$$

By Definition 2.3 HB Chapter 14 p72

Where  $d_0$  is the discrete metric defined on  $\mathbb{R}$

$$d_0(y_1, y_2) = \begin{cases} 0, & \text{if } y_1 = y_2, \\ 1, & \text{if } y_1 \neq y_2. \end{cases}$$

Consider  $y = 4$

$$B_d((0, 0), 1) = \{(x, y) \in \mathbb{R}^2 : d((0, 0), (x, y)) < 1\}$$

$$= \{(x, y) \in \mathbb{R}^2 : |0 - x| + d_0(0, y) < 1\}$$

$$= \{(x, y) \in \mathbb{R}^2 : |x| + d_0(0, y) < 1\}$$

If  $y \neq 0$ , then  $d_0(0, y) = 1$

$$B_d((0, 0), 1) = \{(x, y) \in \mathbb{R}^2 : |x| + 1 < 1\}$$

$$= \{(x, y) \in \mathbb{R}^2 : |x| < 0\}$$

$$= \{(0, y) \in \mathbb{R}^2 : y \neq 0\}$$

If  $y = 0$ , then  $d_0(0, y) = 0$

$$\begin{aligned} B_d((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : |x| + 0 < 1\} \\ &= \{(x, 0) \in \mathbb{R}^2 : |x| < 1\} \\ &= \{(x, 0) \in \mathbb{R}^2 : -1 < x < 1\} \end{aligned}$$

Thus,

$$B_d((0, 0), 1) = \{(x, 0) \in \mathbb{R}^2 : -1 < x < 1\} \cup \{(0, y) \in \mathbb{R}^2 : y \neq 0\}$$